

Ex.No:15 Read and write data from/to files in Python

Programs Date:

Aim: To Implement Read and write data from/to files in Python Programs

Procedure:

1. **Start**
2. Open a file in read or write mode using open().
3. Write data to a file using write() method.
4. Read data from the file using read() or readlines().
5. Append the file.
6. Close the file using close().
7. **End**

Program:

```
# READ AND WRITE DATA FROM / TO FILE IO STREAM
```

```
def writefile(filename, content):
```

```
    with open(filename, "w") as
```

```
        file: file.write(content)
```

```
    print(f"Data written to
```

```
{filename}") def readfile(filename):
```

```
    try:
```

```
        with open(filename, "r") as
```

```
            file: data = file.read()
```

```
    print(f"Data read from {filename}: \n{data}")
```

```
except FileNotFoundError:

    print(f"Error: {filename} not

    found.")

def appendfile(filename, content):

    with open(filename, "a") as file:

        file.write("\n" + content)

    print(f"Data appended to

    {filename}")

# Streams Example: File I/O Operations

filename = "D:\KK\MCADATA.txt"

# Writing to a file (Output Stream)

writefile(filename, "HI MCA STUDENTS, WELCOME TO SRM IST TO

LEARN PYTHON")

# Reading from the file (Input Stream)

readfile(filename)


# Appending new data (Output Stream - Append Mode)

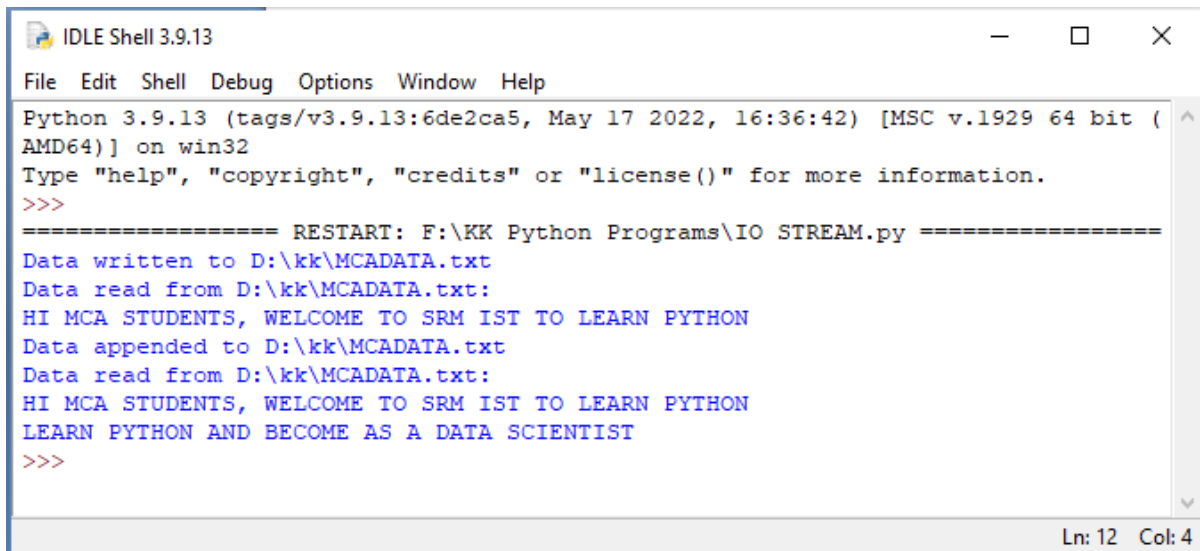
appendfile(filename, "LEARN PYTHON AND BECOME AS A DATA

SCIENTIST")


# Reading updated file (Input Stream)

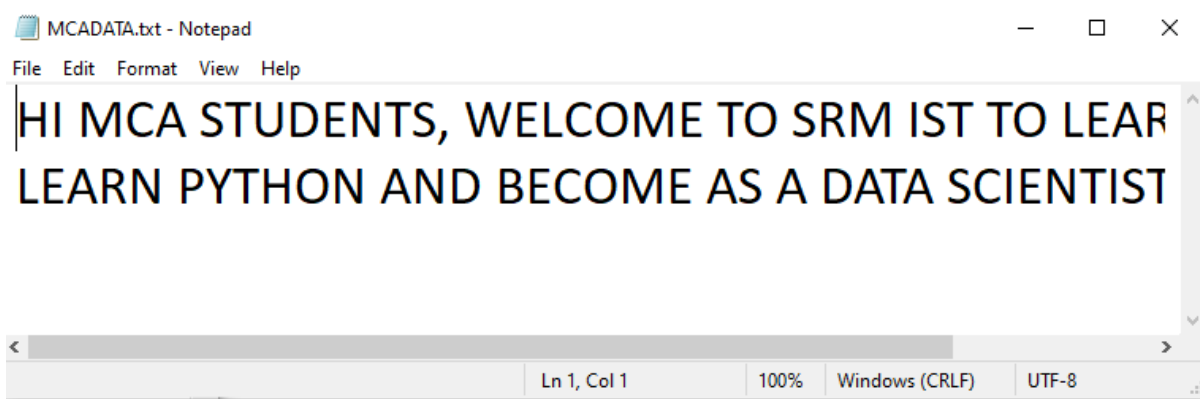
readfile(filename)
```

OUTPUT



```
IDLE Shell 3.9.13
File Edit Shell Debug Options Window Help
Python 3.9.13 (tags/v3.9.13:6de2ca5, May 17 2022, 16:36:42) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: F:\KK Python Programs\IO STREAM.py =====
Data written to D:\kk\MCADATA.txt
Data read from D:\kk\MCADATA.txt:
HI MCA STUDENTS, WELCOME TO SRM IST TO LEARN PYTHON
Data appended to D:\kk\MCADATA.txt
Data read from D:\kk\MCADATA.txt:
HI MCA STUDENTS, WELCOME TO SRM IST TO LEARN PYTHON
LEARN PYTHON AND BECOME AS A DATA SCIENTIST
>>>
Ln: 12 Col: 4
```

MCADATA.txt



```
MCADATA.txt - Notepad
File Edit Format View Help
HI MCA STUDENTS, WELCOME TO SRM IST TO LEARN PYTHON
LEARN PYTHON AND BECOME AS A DATA SCIENTIST
Ln 1, Col 1 100% Windows (CRLF) UTF-8
```

Result:

Thus the program has been executed and verified